

Lecture 19: Multi-Level Page Table

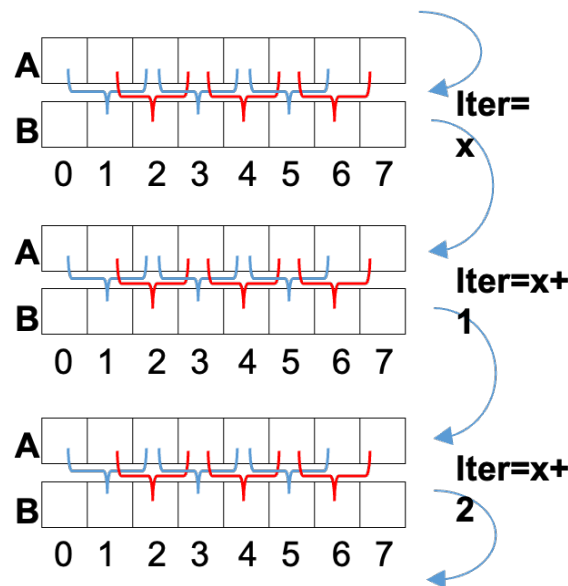
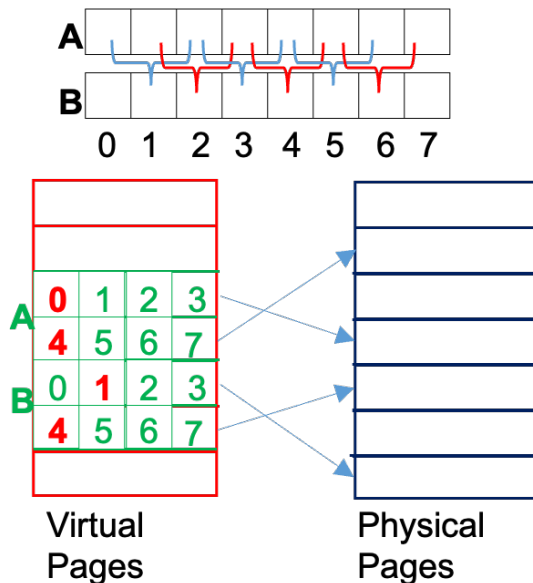
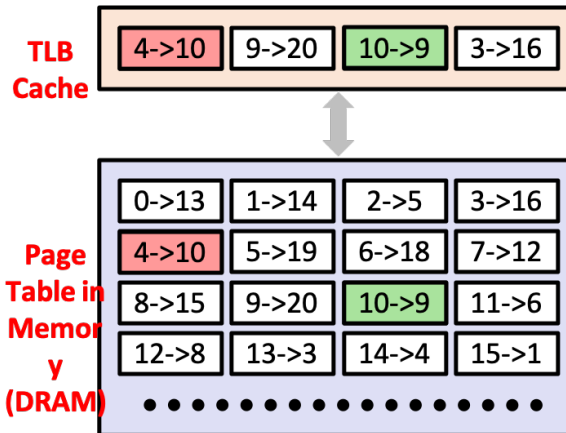
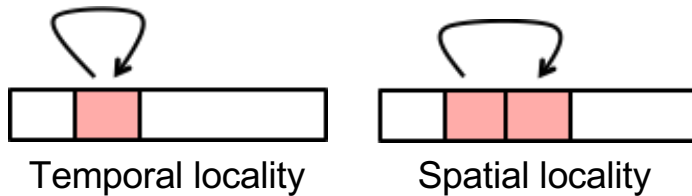
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Last Lecture

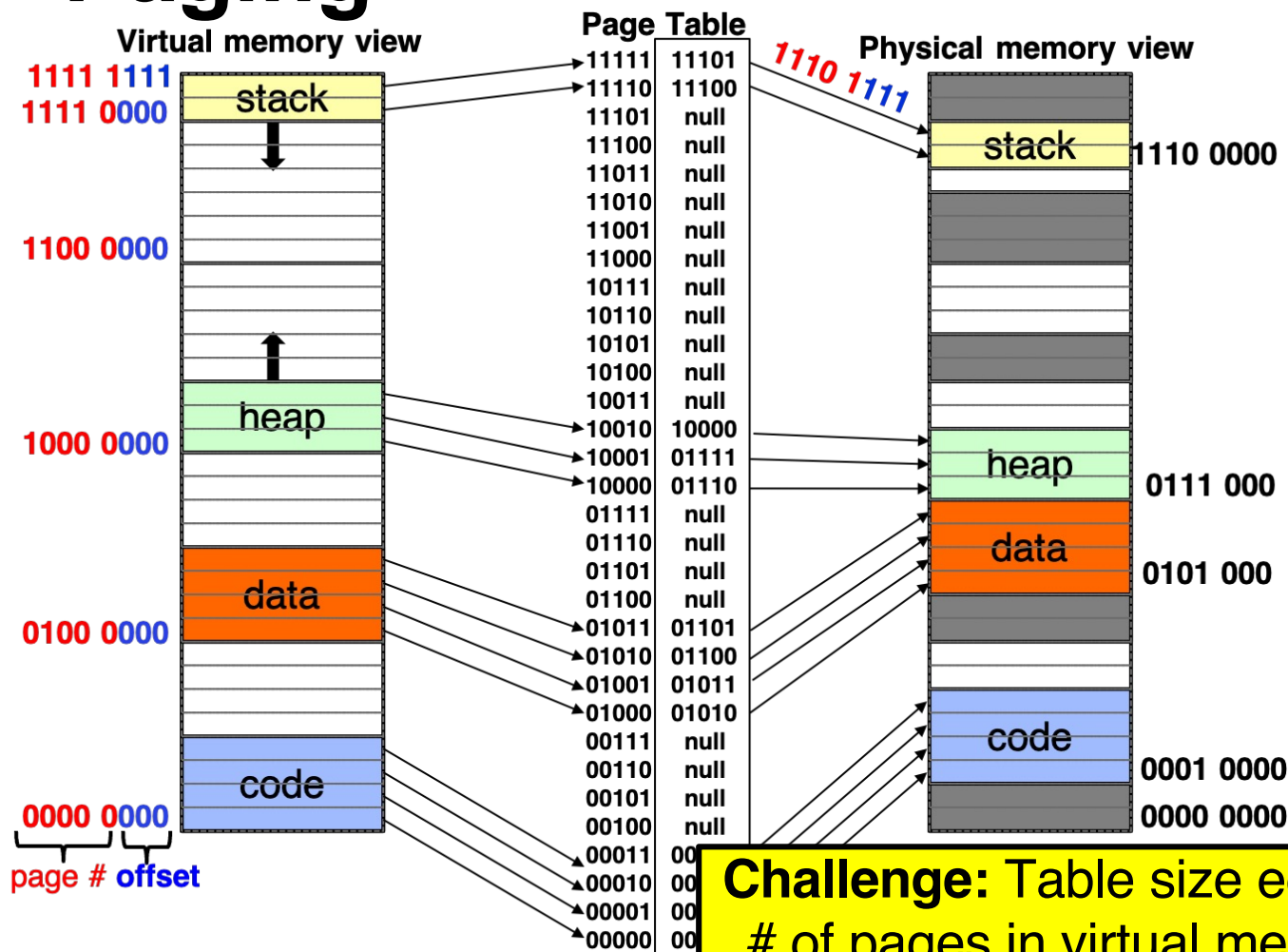


- Time overheads with page table access
 - PT base address in PTBR (1-cycle)
 - Accessing PT index in DRAM (several cycles)
- Translation look aside buffer (TLB)
 - hardware cache per CPU for saving frequently accessed PTE entries (part of MMU)
- Locality helps in caching
 - Temporal locality
 - Spatial locality

Today's Class

- Challenges with simple page table
- Multi-Level Page Table (MLPT)
- MLPT with TLB

Paging



- Recall, process address space is not contiguous
 - Inter-segment memory is not contiguous
 - Only intra-segment virtual memory is contiguous (the physical memory may/may-not be contiguous)
- Page table until now is an **array** where the length of the array will be the total number of VPN being used in the process address space
 - Not all the VPNs will be used as the segments are not contiguous (there could be gaps between them)

Overheads with the Linear Page Table

Discussion for today

- **Time overhead**

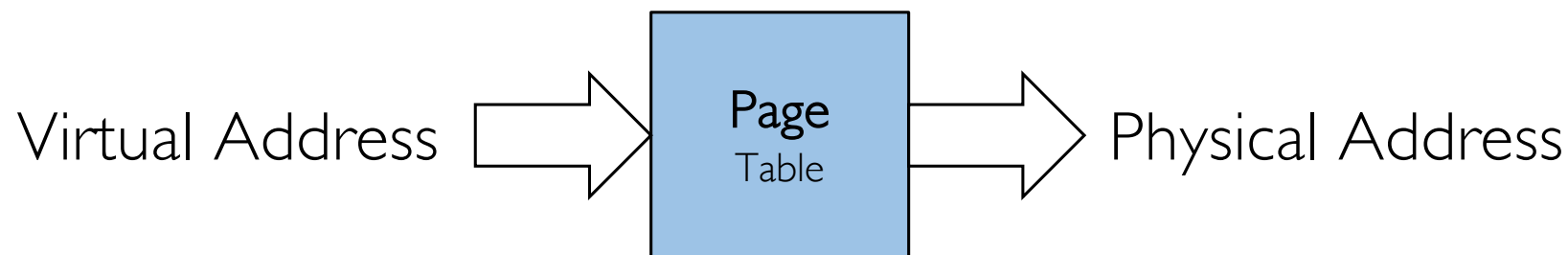
- Covered in last lecture

- **Space overhead (array based implementation)**

- Single entry in PTE on x86 (**32-bit**) = 4 bytes
- Total number of unique VPNs = $2^{32}/4\text{KB} = 2^{32}/2^{12} = 2^{20}$
- Total memory for one page table = $2^{20} \times 4 = \mathbf{4MB}$
- For 100 processes, total memory for the page table = **400MB!**
- On 64-bit: 2^{52} unique VPNs $\times$ 8 bytes = **36 Exa-bytes!**

Challenge: How to Structure a Page Table?

- Page Table is a *map* (function) from VPN to PPN



- Simple (linear) page table corresponds to a *very large* lookup table
 - VPN is index into table, each entry contains PPN
- What other map structures can you think of?
 - Trees?
 - Hash Tables?

A Simple Fix

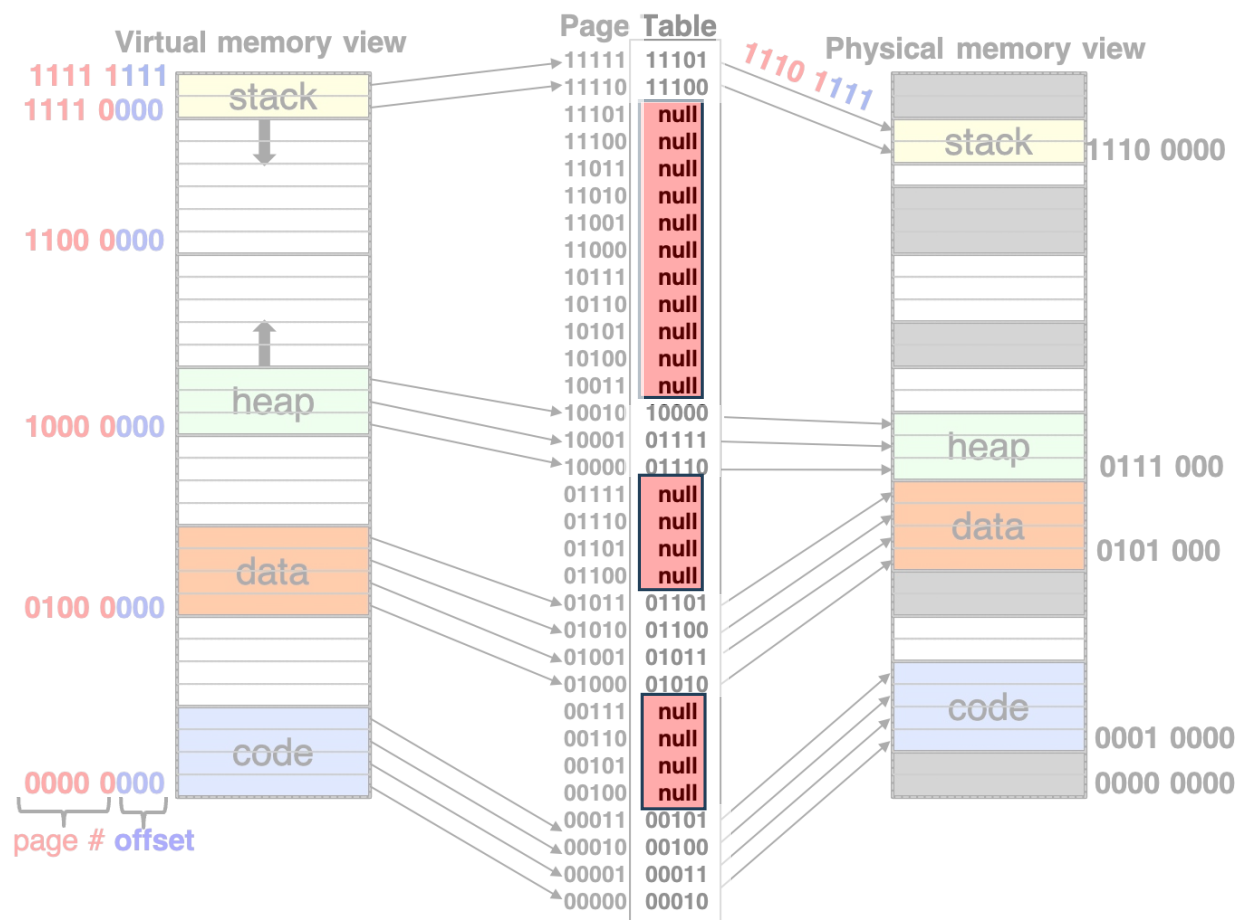
- How about using a page size bigger than 4KB?
- Let's try using a 16KB page size in 32-bit addressing
 - Total unique VPNs are $2^{32}/16\text{KB} = 2^{18}$
 - Space overhead
 - Single entry in PTE on x86 (32-bit) = 4 bytes
 - Total number of unique VPNs = 2^{18}
 - Total memory for one page table = $2^{18} \times 4 = \mathbf{1MB}$
 - We are able to reduce single PT size by a factor of 4!
- Although we are able to alleviate the space overheads by increasing the page size, but do you see any issues in this approach?
 - Internal fragmentation!

Mapping Virtual to Physical Address

● Approaches

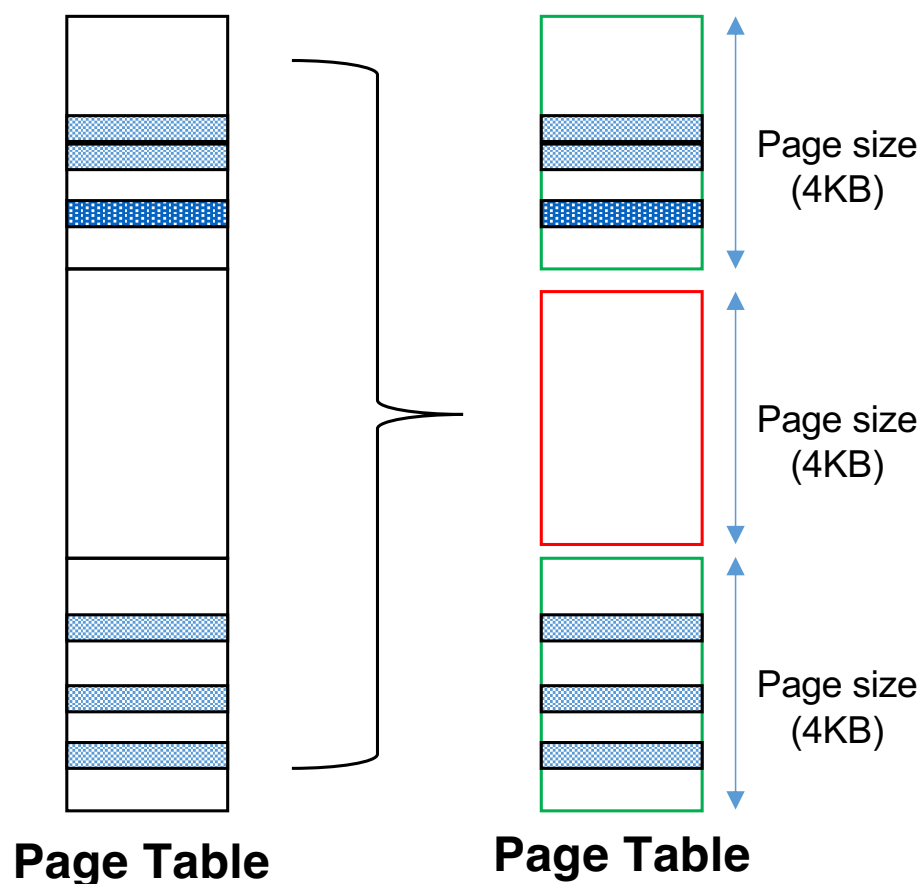
1. Virtual address is same as the physical address
2. Virtual address is not the same as physical address
 - a) Using base & bound for VA to PA mapping
 - b) Using segmentation for VA to PA mapping
 - c) Paging
 - a) Using Page Table (PT) in RAM and having a base register for PT
 - b) Using Page Table (PT) in RAM and having a base register for PT as well as a TLB cache
 - c) **Using Multi-level Page Table**

Reason for Large Page Table



- There are several invalid (“null”) entries in an array-based implementation of the page table
 - Not all indices are going to be valid in the look up table
- We can reduce the size of the page table if we can use a data structure that can help in removing the invalid “null” entries inside the page table
 - **Multi-Level Page Table (MLPT)**

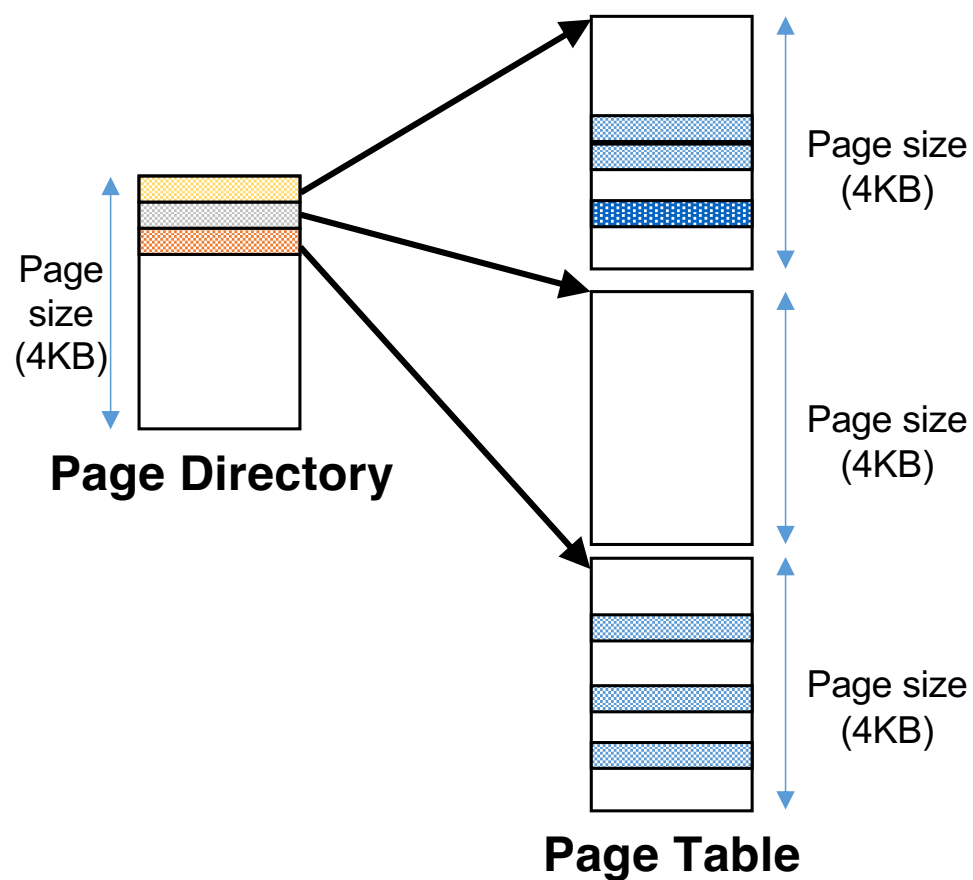
Multi-Level Page Table (MLPT)



● Idea-1

- Split the entire page table into page size entries
- How many entries inside each page size entries?
 - On x86 with 4KB page size, there would be 1024 VPN entries inside each page size blocks (4KB/4bytes)
- Types of “Pages”
 - All invalid (“null”) VPNs
 - At least one valid VPN
- Guess what we can do next?

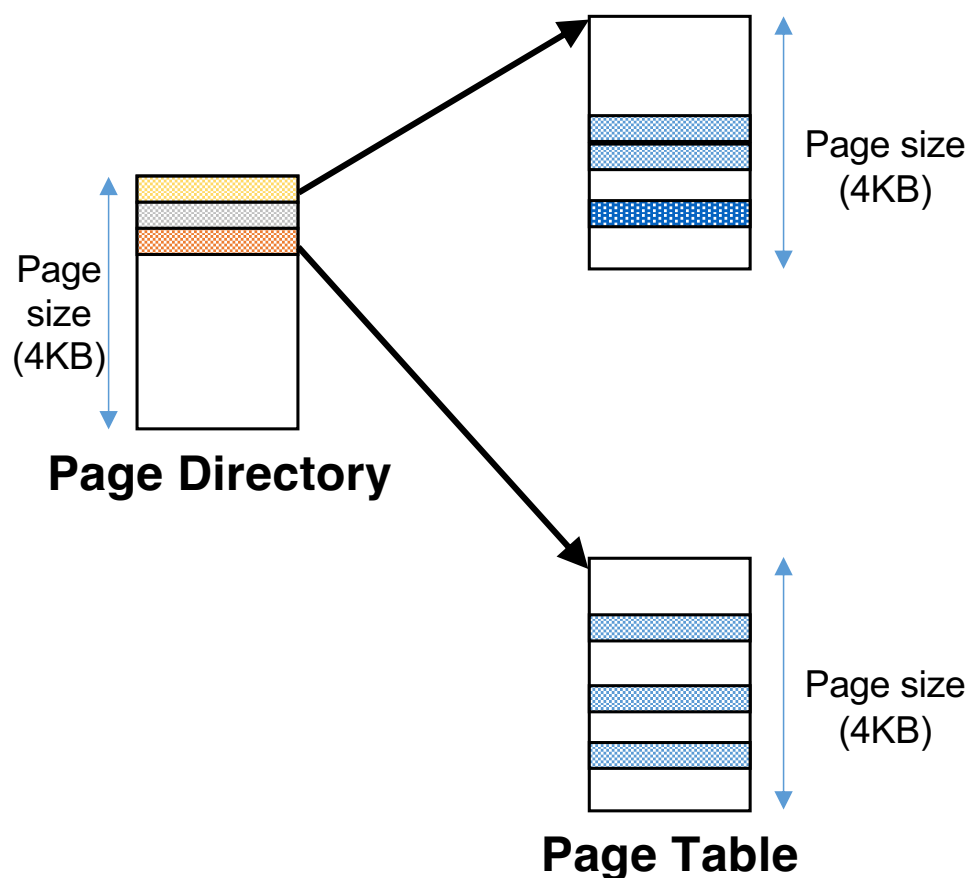
Multi-Level Page Table (MLPT)



● Idea-2

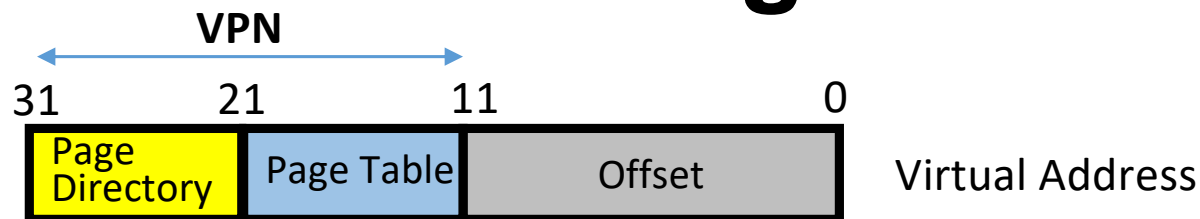
- Create a Page Directory (PD)
 - Per-process array-based data structure
 - Each index in PD points to a page of page table

Multi-Level Page Table (MLPT)



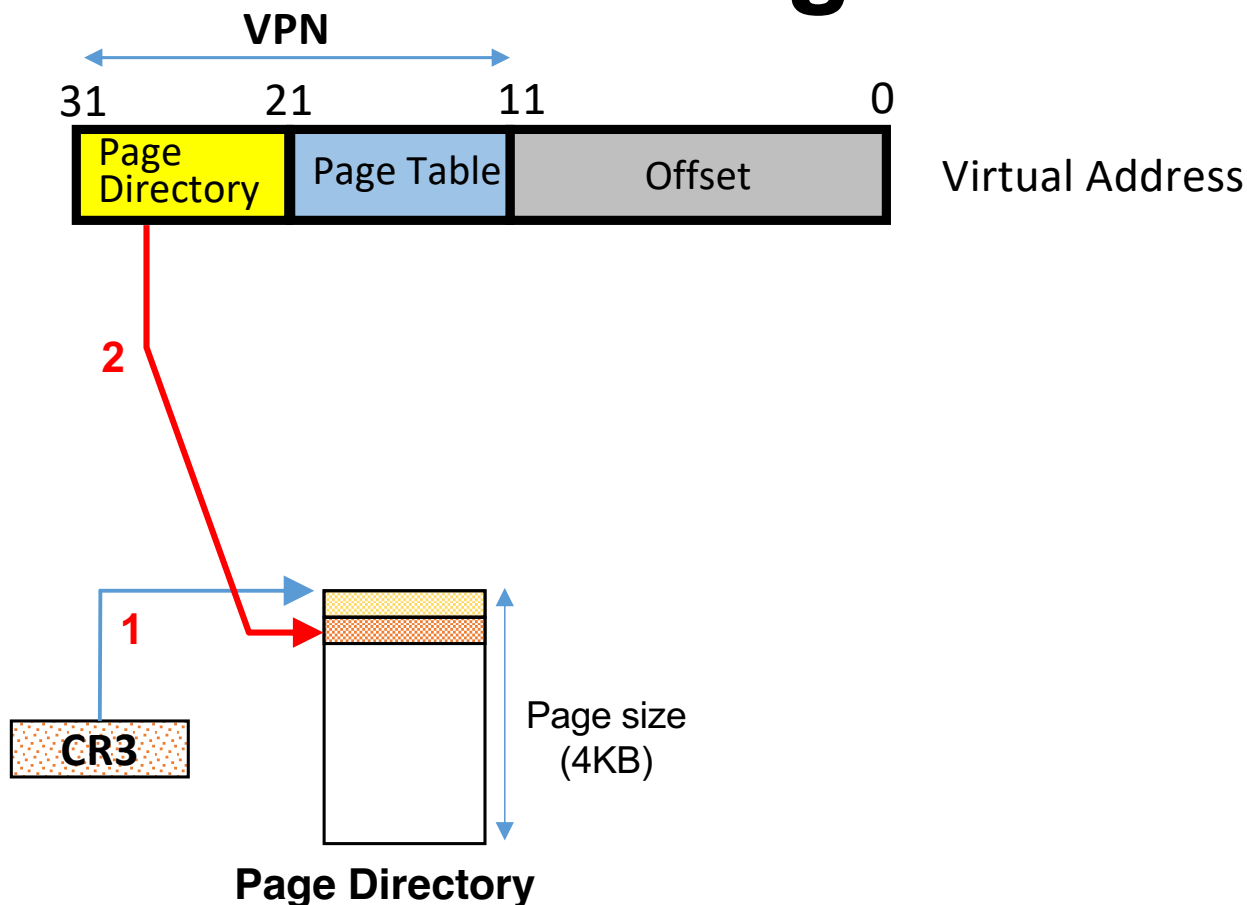
- **Idea-3**
 - Don't allocate a page of page table if there is not a single valid entry inside it
 - PD entry also stores this information apart from the base address of the page of a page table
- This example of MLPT is a two-level page table
- The unused page table in MLPT can easily be swapped to the disk
 - Only the page directory and the page table in use is needed to be on the physical memory
- There could be more than two-levels of page table!
 - **Out of scope of this course**

Two-Level Page Table in x86



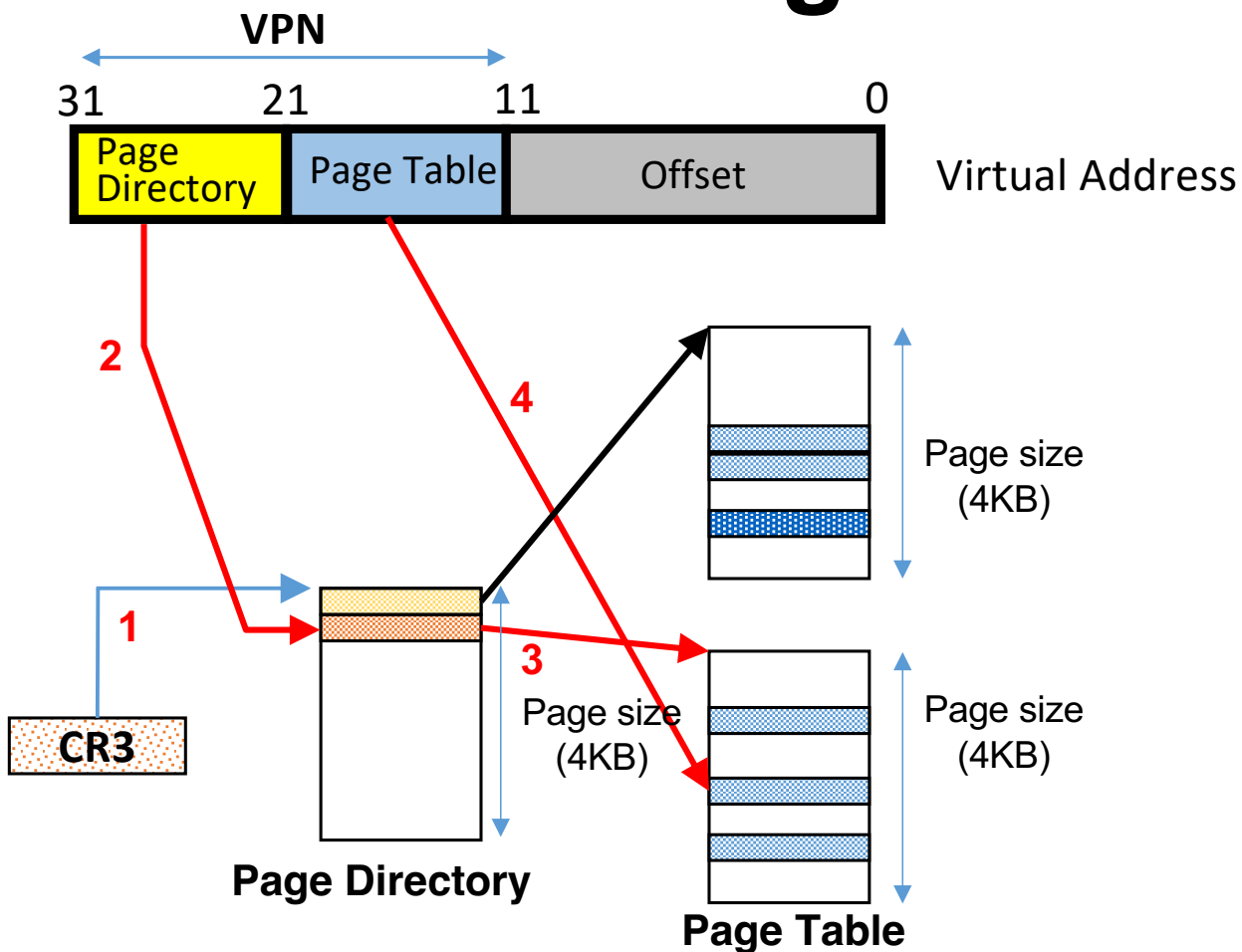
- Virtual address (32-bit) contains three different entries (instead of two that we saw earlier in case of linear page table)
 1. PDIndex – Page Directory index
 - Total number of pages needed to store 2^{20} VPN to PPN mappings = $1024 = 2^{10}$ (each page can contain 2^{10} VPN-PPN mappings)
 - Bits needed to uniquely identify 2^{10} pages = 10
 2. PTIndex – Index into page table corresponding to the above entry inside page directory
 - Total entries in a single page = $1024 (2^{10})$
 - Bits needed = 10
 3. Offset to address in the physical page

Two-Level Page Table in x86



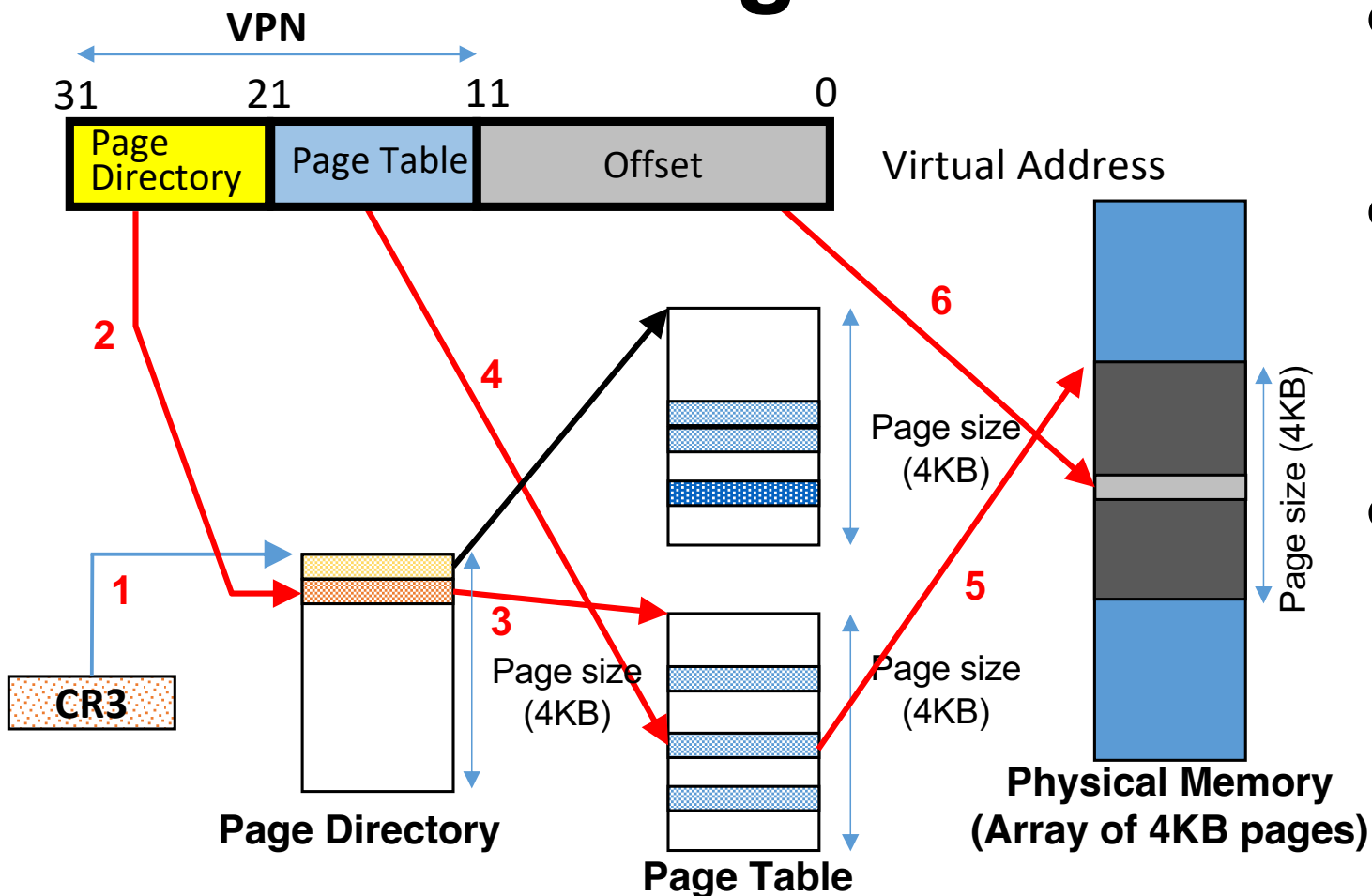
- Control Register (CR3) register (part of MMU) stores the starting address of the Page Directory (PD)
- PD contains 1024 Page Directory Entry (PDEs)
 - Each PDE contains the Page Frame Number and a status bit
 - Status=0 if there is not a single valid entry in the PFN, otherwise status =1 (valid PFN only if status=1)
- Address of the PDE associated with this VPN
 - $\text{PDE} = \text{CR3} + \text{PDIndex} * 4 \text{ bytes}$

Two-Level Page Table in x86



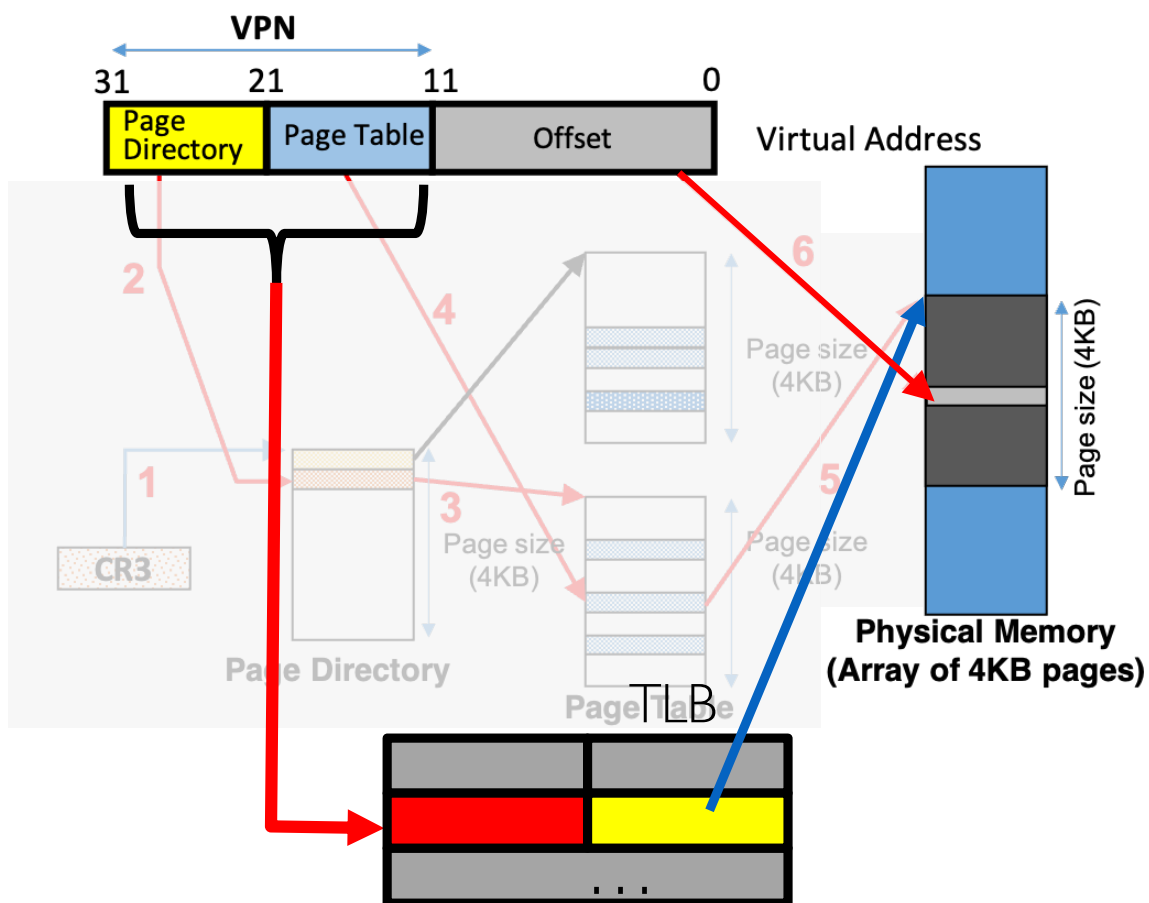
- Page Table Entry (PTE) can be now found from the PDE
 - $\text{PTE} = \text{PDE} + (\text{PTIndex} * 4 \text{ bytes})$

Two-Level Page Table in x86



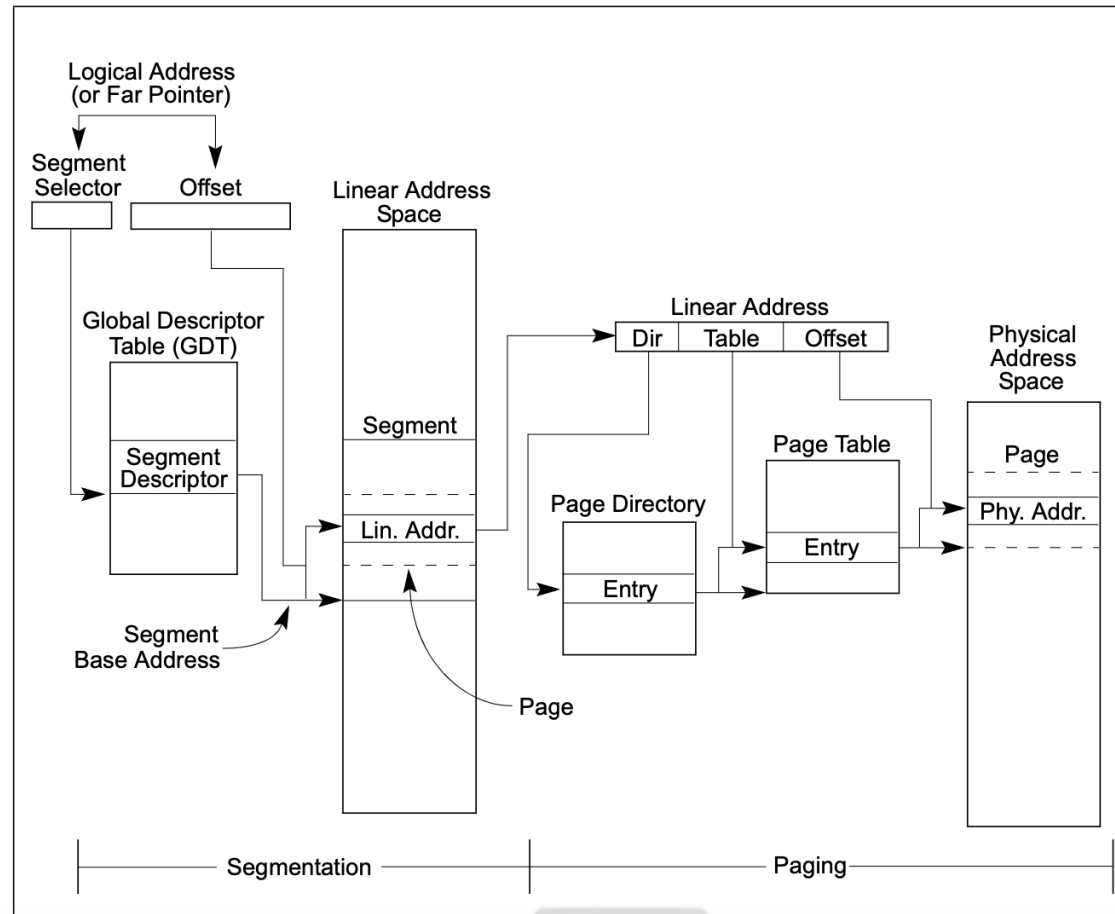
- Physical Page Frame Number (PFN) can be found from the PTE
- The physical address is obtained by adding the “offset” into the starting address of PFN
- What would happen when considering TLB?
 - None of these complex lookups are required if TLB contains the VPN->PPN

MLPT Along with TLB



- Look ups are required only in case of TLB misses
- Access latency significantly low when TLB hits
 - See Lecture-18

Memory Management in Intel x86

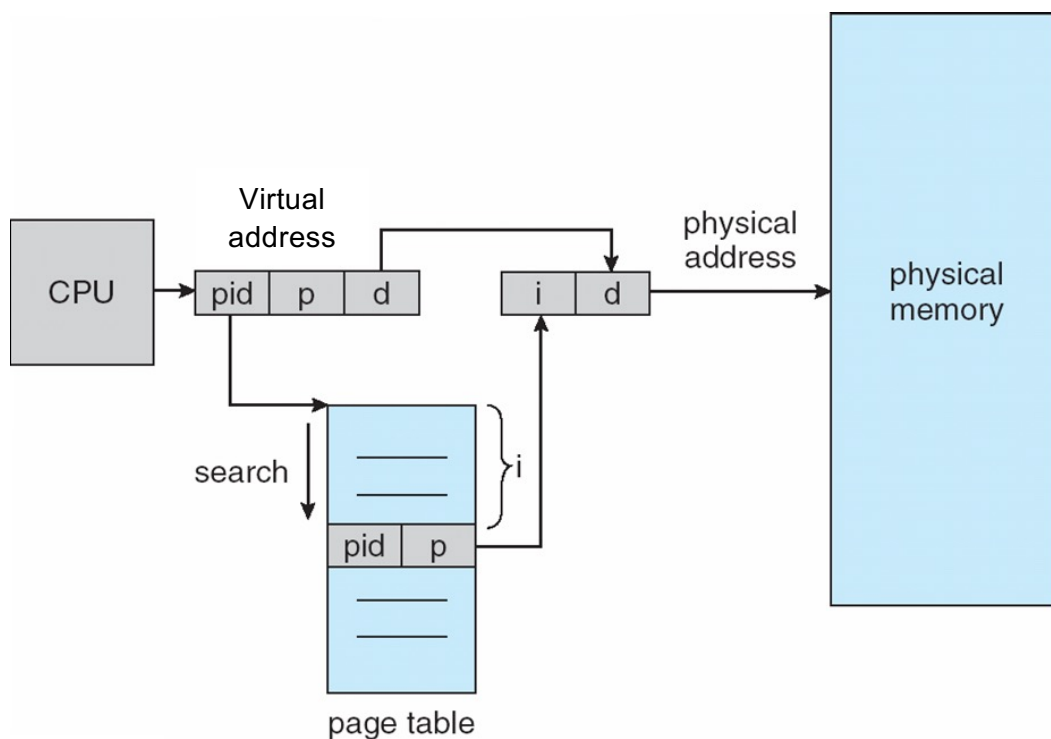


Mapping Virtual to Physical Address

● Approaches

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 - d) **Using Inverted Page Table**

Inverted Page Table



Picture source: <https://www.os-book.com/OS10/>

- Issues with MLPT
 - Easy implementation but many page tables
- Inverted Page Table
 - Size is independent of the virtual address space, and is directly related to the amount of physical memory
 - Each entry points to a page in the physical memory
 - Each entry corresponds to the VPN ("p") and process ("pid") that owns that page
 - The tuple (pid, p) inside a virtual address is used as a hash key to find the entry in the page table
 - Single page table contains entries for all the processes
 - Complex hash function!
 - It is called as inverted page table, as total entries is equal to the total number of PPN
 - In contrast, the page tables we have seen so far (a.k.a. forward page tables) the entries are according to the VPN

Next Lecture

- Demand paging
 - Last remaining topic in virtualization of memory